



UNIVERSITY OF THE PUNJAB

Eight Semester 2014
Examination B.S. 4 Years Programme

Roll No. _____

TIME ALLOWED: 2 hrs. & 30 min.
MAX. MARKS: 50

PAPER: Time Series Analysis-II
Course Code: STAT-413

Attempt this Paper on Separate Answer Sheet provided

Notes: Attempt all Questions.

Q.No.1. Describe the following:

- Iterative procedure of finding the Least Squares estimates of and MA (1) process with non-zero means. ✓ 194
- Importance of diagnostic checking stage, in Time series model building.
- Revision of forecasts and its importance in long term forecasting.
- Extrapolation of trend curves.
- ✓ Akaike's Information Criterion (AIC), for model selection.

Q.No.3. The following table gives the auto-correlations for residuals of an ARMA (1,1) model fitted to a series of 100 observations. (5)

Lag	1	2	3	4	5	6	7	8	9	10
$r_k(\hat{Z})$	0.03	0.05	-0.01	0.12	0.06	-0.08	0.15	0.10	0.09	0.05

With the help of Box-Pierce statistic check the appropriateness of the fitted model

Q.No.4. Find the MMSE forecast of X_{n+h} for an AR (1) process, with zero mean, using MA representation approach. (4)

Q.No.5. Given an AR (1) process
(1 - 0.7B)(X_t - 10) = Z_t

We have X_t; t = 1, 2, ..., 100 such that

$$x_{97} = 10.6, \quad x_{99} = 10, \quad x_{100} = 9.9$$

Forecast $x_{101}, x_{102}, x_{103}$.

Obtain 95% confidence limits x_{101}, x_{102}

Suppose that x_{101} turns out to be 8.8, update x_{102}, x_{103}

$$x_{101} = 8.8$$

Q.No.6. If $\{X_t\}$ follows an AR (1) process with the assumption that $Z_t \sim N(0, \sigma_z^2)$

Then show that

$$\ln(L) = \left[\text{const} - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} \ln(1 - \phi_1^2) - \frac{1}{2\sigma_z^2} (A - 2\phi_1 B - \phi_1^2 C) \right]$$

Where,

$$A = \sum_{t=1}^n X_t^2; \quad B = \sum_{t=2}^n X_t X_{t-1}; \quad C = \sum_{t=2}^{n-1} X_t^2$$

Also show that MLE of σ_z^2 is:

$$\hat{\sigma}_z^2 = \frac{A - 2\phi_1 B - \phi_1^2 C}{n} \quad \text{And approximate MLE of } \phi_1 \text{ is } \hat{\phi}_1 = \frac{B}{C}$$

AIC model selection.
The Akaike information criterion (AIC) is an estimator of the quality of statistical models for a given



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Roll No.

PAPER: Time Series Analysis II
Course Code: STAT 413

TIME ALLOWED: 2 hrs. & 30 mins
MAX MARKS: 50

Attempt this Paper on Separate Answer Sheet provided

Note: Attempt all questions.

Q.No.2. Answer the following:

- Explain residual analysis as a tool for diagnostic checking in time series model building. (20) P 194
- Explain why it is difficult to estimate parameters of MA process also describe the procedure of estimating MA process by using dual estimation method. 193
- Discuss akaike's information criteria (AIC), for model selection.
- Describe seasonal ARIMA process. Write down the model for SARIMA (1,0,0)_X(0,1,1)₁₂. 231 - 232

Q.No.3. Given an AR (2) process,

$$X_t = \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + Z_t; \text{ where } Z_t \sim N(0, \sigma_z^2)$$

Then show that

$$\ln(L) = \text{const} - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} \ln |M_2| - \frac{1}{2\sigma_z^2} S(\varphi_1, \varphi_2)$$

Where,

$$M_2 = \begin{bmatrix} 1 & \varphi_1 & \varphi_2 \\ \varphi_1 & 1 & \varphi_1 \\ \varphi_2 & \varphi_1 & 1 \end{bmatrix} \quad C = \begin{bmatrix} X_1^2 \\ X_2^2 \\ \vdots \\ X_n^2 \end{bmatrix}$$

$$\text{If } Y_0 = \frac{(1 - \varphi_2)\sigma_z^2}{(1 + \varphi_2)[(1 - \varphi_2)^2 - \varphi_1^2]} \quad \& \quad Y_1 = \frac{\varphi_1 Y_0}{(1 - \varphi_2)}$$

Then evaluate M_2 and $S(\varphi_1, \varphi_2)$, then find MLE of σ_z^2 and suggest method of MLEs of φ_1 & φ_2 . 174

Q.No.4.

- Explain the meaning and importance of updating forecasts. Derive the rule to update the forecast previously made with origin at "n" using the information X_{n+1} . (2x6) 228
- Find the MMSE forecast of X_{n+1} for an AR(1) process, with zero mean using MA representation approach. Also find MMS forecast error for the process and discuss the structure of MMS forecast error. 205

Q.No.5. For the model,

$$(1 - B)(1 - 0.2B)X_t = (1 - 0.5B)Z_t$$

Find forecast for one and two steps ahead and show that

$$X_n(l) = 1.2X_n(l-1) - 0.2X_n(l-2); \text{ for } l \geq 3$$

If $\sigma_z^2 = 0.10$, also find $\text{Var}[e_n(1)]$ & $\text{Var}[e_n(2)]$ P 222

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Attempt this Paper on Separate Answer Sheet provided.

Note: Attempt all questions.

Question No.2: Answer the following

13/ a) Shortly describe the four stages involved in time series model building.

b) Define forecasting; write a note on extrapolation of trend curves.

c) If there are more than one appropriate models for a time series data, on what grounds would you select the best appropriate among the candidate models.

d) Show that for ARIMA (1,1,1) model, the forecast function for lead time "l" and origin at "n" is

$$X_n(l) = X_n + \frac{\varphi(1-\varphi^l)}{(1-\varphi)}(X_n - X_{n-1}) + \frac{\theta(1-\varphi^l)}{(1-\varphi)}Z_n \quad 221$$

Question No. 3. Show that the Minimum Mean Square Error forecast with origin at "n" and lead time "l" is given by: (10)

$$X_n(l) = \frac{1}{\Psi(B)} \left[\frac{\Psi(B)}{B^l} \right] + X_n \quad 201$$

Where $\Psi(B)$ the polynomial of the process in moving average representation.

Question No. 4. a) Following table gives the autocorrelations for residuals of an ARMA(1,1) model fitted to a series of 100 observations.

Lag	1	2	3	4	5	6	7	8
$r_k(Z_t)$	0.03	0.05	-0.01	0.12	0.06	-0.08	0.15	0.10

With the help of Box-Pierce statistics, check the appropriateness of the fitted model.

b) Explain why it is difficult to estimate the parameters of MA process. Describe the procedure of finding the iterative least squares estimates of MA(1) process given by

$$X_t = \mu + \theta Z_{t-1} + Z_t$$

✓ ✓ ✓
✓ 191-192
190+191+192

Question No. 5. If $\{X_t\}$ follows an AR(2) Process with zero mean, then find the least



UNIVERSITY OF THE PUNJAB

Eighth Semester - 2017

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Time Series Analysis-II
Course Code: STAT-413

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Note: Attempt all questions.

Question No.2. Answer the following.

(4 × 5 = 20)

- a) Discuss the role of ACF & PACF in identification of a time series model.
- b) Consider the $SARIMA(1, 0, 0) \times (0, 1, 1)_{12}$ Model and show that
$$X_n(2) = X_{n-10} + \varphi^2(X_n - X_{n-12}) + \theta\varphi Z_{n-11} + \theta Z_{n-10}$$
- c) Describe Portmanteau Lack of fit tests..
- d) Find the MMSE forecast of X_{n+l} for an AR(1) process, with zero mean, using MA representation approach

Question No. 3. Given an AR(2) process, $X_t = \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + Z_t$,

(10)

Find MLE of σ_z^2 , also find approximate MLEs of φ_1 and φ_2 .

Question No. 4. Explain the meaning and importance of updating forecasts. Derive the

(10)

rule to update the forecast previously made with origin at "n" using the information X_{n+1} .

Question No. 5. For the model $(1 - B)(1 - 0.2B)X_t = (1 - 0.5B)Z_t$

(10)

Find forecast for one and two steps ahead and show that

$$X_n(l) = 1.2X_n(l-1) - 0.2X_n(l-2), \text{ for } l \geq 3$$

If $Z_n = 1$, $X_n = 4$, $X_{n-1} = 3$ & $\sigma_z^2 = 0.10$.

Also show that

$$X_n(2) = 3.64 \text{ and } \text{Var}[e_n(1)] = 0.10.$$





UNIVERSITY OF THE PUNJAB

Eighth Semester – 2018

Examination: B.S. 4 Years Program

Roll No. 010606

PAPER: Time Series Analysis-II

Course Code: STAT-413 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2 Answer the following:

(4 x 5)

- Describe the four stages of Box-Jenkins model building approach.
- Describe briefly the iterative procedure for least squares estimation of an MA(1) process.
- Show that the forecast errors for lead time 1 with different forecast origins are always uncorrelated.
- Derive the minimum mean squares error forecast function for an AR(1) process.

Q.3 If $\{Y_t\}_{t=1}^n$ follows an AR(2) process $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$ where $\{Z_t\}$ is independently and normally distributed process with zero mean and finite variance σ_z^2 then show that log-likelihood function is (10)

$$\ln L = \text{const.} - \frac{n}{2} \ln \sigma_z^2 - \frac{1}{2} \ln M_2 - \frac{1}{2\sigma_z^2} \left(\sum_{t=3}^n \left(Y_t - \sum_{i=1}^2 \phi_i Y_{t-i} \right)^2 + Y^T M_2 Y \right)$$

where $Y = (Y_1, Y_2) \sim N_2(0, \Sigma_2)$ and $M_2 = \sigma_z^2 \Sigma_2^{-1}$.

Q.4 Show that for an ARIMA(1,1,1) process: $(1 - \phi B) \nabla Y_t = (1 + \theta B) Z_t$, the forecast at origin t with lead time l is given by (10)

$$Y_t(l) = Y_t + \frac{\phi(1 - \phi^l)}{1 - \phi} (Y_t - Y_{t-1}) + \frac{\theta(1 - \phi^l)}{1 - \phi} Z_t$$

Q.5 An AR(1) model $(1 + 0.5B)(Y_t - 50) = Z_t$ is fitted to an observed time series of 80 observations. Now a practitioner wants to use this model to forecast Y_{81} and Y_{82} . Compute the desired MMSE forecasts and 95% forecast interval if $Y_{79} = 59, Y_{80} = 52, \sigma_z^2 = 4$. Update the forecast of Y_{82} if it is observed that $Y_{81} = 62$. (10)



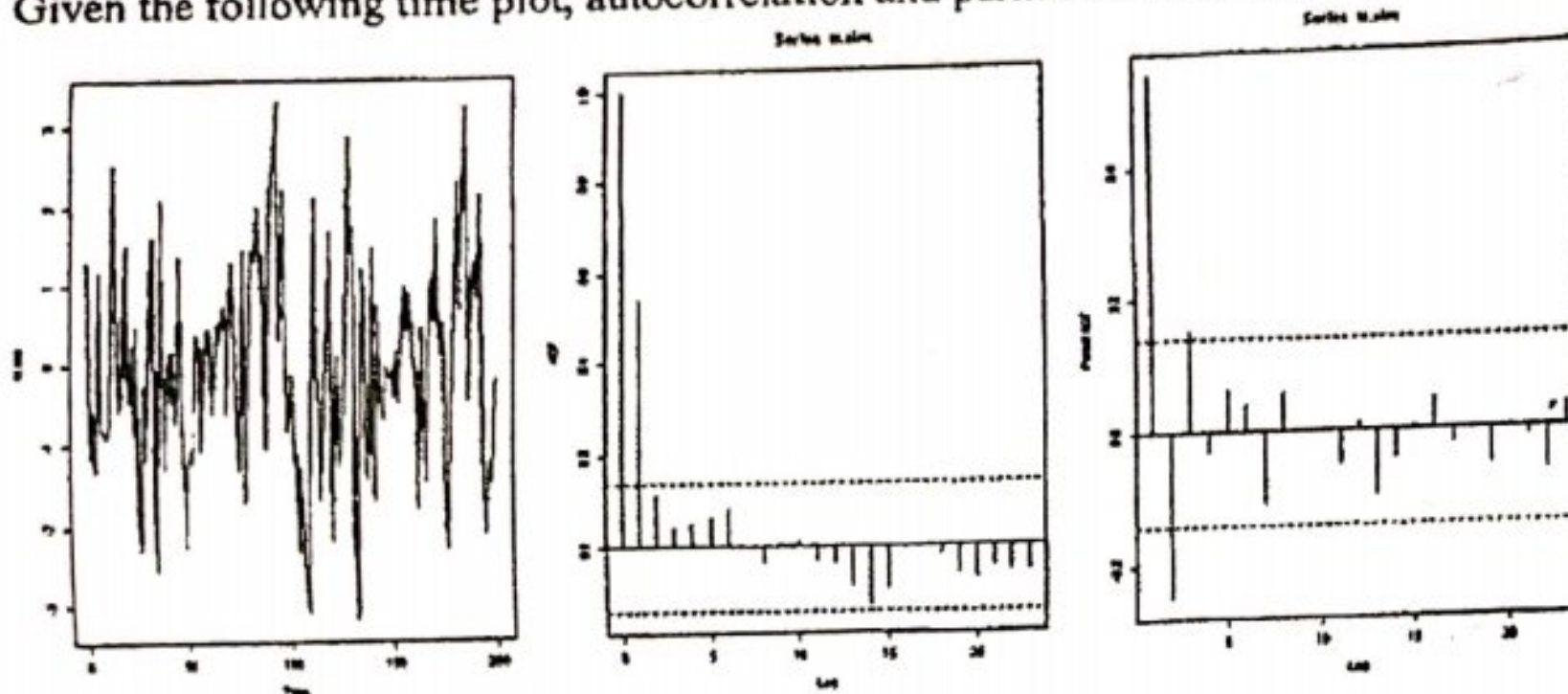


ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Note:-Attempt ALL the questions.

Q.2 Given the following time plot, autocorrelation and partial autocorrelation functions:

[6]



- Discuss the graphs behavior.
- Identify the candidate ARMA model(s).

Q.3 Given the following information, estimate the parameters of AR(2) model.

[6]

$$\bar{Y} = 0.21, r_1 = 0.87, r_2 = 0.83$$

Q.4 Following are the autocorrelations obtained for the residuals by fitting an ARMA(2,1) process to an observed time series with $n = 200$.

[8]

k	1	2	3	4	5
r_k	0.04	-0.13	0.25	-0.01	0.09

State the hypothesis and test the goodness of fitted model using a portmanteau test for $m = 5$. Use 5% level of significance.

Q.5 Given the following AR(2) process

[10]

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$$

Derive the loss function for the Maximum likelihood estimation of AR parameters. Also obtain the MLEs (approximate) of ϕ_1, ϕ_2 and σ_z^2 .

Q.6 Show that the minimum mean squared error forecast with origin at n and lead time l is for an AR(1) process $Y_t = \phi Y_{t-1} + Z_t$ is given by

[10]

$$Y_n(l) = \phi^l Y_n$$

with forecast error variance

$$\text{var}(e_n(l)) = \sigma_z^2 \left(\frac{1 - \phi^{2l}}{1 - \phi^2} \right)$$

Q.7 An AR(1) model $Y_t - 53 = 0.65(Y_{t-1} - 53) + z_t$, is fitted to an observed time series of

[10]

100 observations. Now a practitioner wants to use this model to forecast Y_{101} and Y_{102} .

- Compute the MMSE forecasts of Y_{101} and Y_{102} , if $Y_{99} = 52.7, Y_{100} = 53.5$ using (a) $\psi(B)$ polynomial. (b) $\pi(B)$ polynomial.
- Suppose that Y_{101} comes out to be 51.3, update the forecast of Y_{102} by shifting forecast origin to $t = 101$.

$$Y_{t+2} = 53 + 0.65(Y_{t+1} - 53) + z_{t+2}$$



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

(4×5=20)

Q.2. Answer the following short questions.

- Discuss the importance of sample ACF & PACF in time series model building.
- Explain the meaning and importance of updating forecasts.
- Describe the procedure of finding the iterative least squares estimates of MA(1) process given by, $X_t = \mu + \theta Z_{t-1} + Z_t$
- If $\{X_t\}$ follows an AR(2) Process with zero mean, then find the Yule-walker estimates of the parameters of the model.

Answer the following questions.

Question No. 3. Following table gives the autocorrelations for residuals of an ARMA(1,1) model fitted to a series of 100 observations. (08)

Lag	1	2	3	4	5	6	7	8
$r_k(Z_t)$	0.03	0.05	-0.01	00.12	0.06	-0.08	0.15	0.10

With the help of Ljung-Box-Pierce statistic, check the appropriateness of the fitted model.

Question No. 4. Given an AR(1) process, $X_t = \phi_1 X_{t-1} + Z_t$, Find MLE of σ_z^2 , also find approximate MLE of ϕ_1 . (10)

Question No. 5. Given an AR(1) process (12)

$$(1 - 0.7B)(X_t - 10) = Z_t$$

We have $X_t: t = 1, 2, \dots, 100$ Such that

$$X_{97} = 10.6, X_{98} = 10, X_{99} = 10, X_{100} = 9.9 \quad \& \quad \sigma_z^2 = 0.21$$

- Forecast $X_{101}, X_{102}, X_{103}$.
- Obtain 95% confidence limits for X_{101}, X_{102} .
- Suppose that X_{101} turns out to be 8.8, update X_{102}, X_{103} .



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following short questions. (4x5=20)

- Describe the Model-Building stages suggested by Box and Jenkins.
- What is meant by Residuals analysis? Describe its role in model diagnostic checking.
- What is a portmanteau test? Describe the null hypothesis and use of portmanteau test.
- Describe the forecasts errors and its role in updating the forecasts

Answer the following questions.

(3x10=30)

Q3.

(10)

If $\{Y_t\}_{t=1}^n$ follows an AR(2) process $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$ where $\{Z_t\}$ is independently and normally distributed process with zero mean and finite variance σ_z^2 then show that log-likelihood function is

$$\ln L = \text{const.} - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} \ln |M_2| - \frac{1}{2\sigma_z^2} \left(\sum_{t=1}^n \left(Y_t - \sum_{i=1}^2 \phi_i Y_{t-i} \right)^2 + X^T M_2 X \right)$$

where $(Y_1, Y_2) \sim N_2(0, \Sigma_2)$ and $M_2 = \sigma_z^2 \Sigma_2^{-1}$. Also suggest procedure to find the approximate maximum likelihood estimates of AR parameters.

Q4.

(10)

Following are the autocorrelations obtained for the residuals by fitting an ARMA(1,2) process to an observed time series with $n = 50$.

k	1	2	3	4	5	6	7	8	9	10
r_k	-0.15	0.08	-0.05	0.01	0.03	0.01	0.05	-0.04	0.01	0.02

State the null hypothesis for the goodness-of-fit testing of fitted model. Test the hypothesis using the Ljung-Box portmanteau test with $m = 5$. What are your recommendations about the fitted model? Use 5% level of significance.

Q5.

(10)

An AR(1) model $(1 - 0.7B)(X_t - 50) = Z_t$ is fitted to an observed time series of 100 observations. Now a practitioner wants to use this model to forecast X_{101} and X_{102} .

- Compute the desired MMSE forecasts if $X_{99} = 49, X_{100} = 52$.
- Obtain 95% forecast limits for X_{101} , if $\sigma_z^2 = 5$.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – Spring 2022

Paper: Time Series Analysis-II

Course Code: STAT-413

Roll No. 226472.....

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answers to the following questions. (6x5=30)

- Discuss Box and Jenkins model building stages.
- Explain Portmanteau tests.
- Define model selection criteria AIC and BIC.
- Write a short note on simple exponential smoothing.
- Discuss different forecasting procedures.
- Define Backcasting.

Write detailed answers to the following questions. (3x10=30)

Q.2. Find the maximum likelihood estimates of ϕ_1 and ϕ_2 for the given AR(2) model.

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

Where $Z_t \sim N(0, \sigma_z^2)$

Q.3. Find the minimum mean square error forecast of X_{n+l} for AR(1) process with zero mean using MA representation. Also find its mean and variance.

Q.4. i) Suppose the Correlogram of the time series of 100 observations has

k	1	2	3	4	5	6	7	8	9	10
r_k	0.31	0.37	-0.05	0.06	-0.21	0.11	0.08	0.05	0.12	-0.01

Suggest an appropriate ARMA model. Calculate 2 sigma confidence limits for autocorrelation.

ii) A computer generates a series of 400 observations that are supposed to be random. First 10 auto-correlations of the time series are given as:

k	1	2	3	4	5	6	7	8	9	10
r_k	0.02	0.05	-0.09	0.08	-0.02	0	0.12	0.06	0.02	-0.08

Is there any evidence for non-randomness?

$$e_n(l) = y_{n+l} - y_n(l)$$



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – Spring 2024

Paper: Time Series Analysis-II

Course Code: STAT-413

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (5x6=30)

- Define residual analysis and why it is important?
- Write a short note on over-parameterization.
- Define Bayesian information criteria for model selection. Also show its approximation for time series models.
- Discuss the behaviour of ACF and PACF of MA(q) and AR(p) process.
- Discuss the importance of forecasting.

Answer the following questions. (3x10=30)

- Use Least square estimation procedure to find out the parameters of AR(2) process with zero mean.
- The first eight values of ACF and PACF of 60 observations on a quarterly economic index and of the first differences are given below:

lag		1	2	3	4	5	6	7	8
X_t	r_k	0.95	0.91	0.87	0.82	0.79	0.74	0.70	0.67
	φ_{kk}	0.95	0.04	-0.05	0.07	0	0.07	-0.04	-0.02
ΔX_t	r_k	0.42	0.20	0.12	0.05	-0.02	-0.05	-0.01	0.03
	φ_{kk}	0.42	0.08	0.06	0.03	-0.05	-0.06	-0.04	-0.02

- Plot the sample ACF and PACF with 95% confidence limits.
- Identify the lags at which sample ACF and PACF is significant.
- Identify the model based on given data.

Q.4. Show that for an ARIMA (1,1,1) process; $(1 - \phi B) \nabla X_t = (1 + \theta B) Z_t$, the forecast at origin t with lead time l is given by:

$$X_n(l) = X_n + \frac{\phi(1-\phi^l)}{1-\phi} (X_n - X_{n-1}) + \frac{\theta(1-\phi^l)}{1-\phi} Z_n$$